ChatGPT

Q1-Readiness Review of Bagged Fuzzy TOPSIS and the Reliability-Aware Variant

Executive summary

Your submitted report proposes an ensemble-style extension of fuzzy TOPSIS in which repeated resampling of decision makers (DMs) produces many fuzzy TOPSIS runs whose outputs are aggregated (e.g., by voting/Borda or by aggregating closeness coefficients), with an "enhanced" variant that learns or assigns DM reliability and uses it to bias sampling/aggregation. filecite turn0file0 The core motivation—robustness to noisy, biased, or adversarial DM inputs—is well aligned with current interest in robustness and stability in MCDM, and ensemble ideas in MCDM are recognised as a small but growing niche. [1]

However, as written today, the work is **not yet Q1-journal worthy**: the novelty claim is vulnerable because "bootstrap/Monte Carlo + (fuzzy) TOPSIS" and "DM weighting/reliability in group TOPSIS" both have clear prior art (near-misses that a reviewer will cite), and the manuscript lacks the theoretical framing, experimental breadth, statistical testing, and reproducibility packaging expected for Q1 outlets. For example, prior work already uses bootstrap sampling to generate **distributions of fuzzy TOPSIS rankings** in large-DM settings, which is conceptually close to DM-bagging, even if the problem setting differs. [2] In parallel, robust fuzzy TOPSIS under Monte Carlo sampling has been published for group decision contexts. [3]

To reach Q1 standards, you should reposition the contribution as a **robust group fuzzy TOPSIS framework under contamination/adversarial models**, explicitly differentiate from bootstrap-for-uncertainty papers, add formal definitions and at least 1–2 theoretical claims (variance/stability and a robustness or breakdown-style argument), and execute a rigorous experimental protocol (multiple real datasets + controlled synthetic corruption + statistical tests + ablations + reproducible code/data). [4]

Novelty and literature coverage

Assessment: Weak (but salvageable with repositioning).
Your report argues for novelty largely by combining "bagging" ideas with fuzzy TOPSIS and then adding reliability-aware sampling/aggregation. filecite turn0file0 In the current literature, reviewers can reasonably respond: **(i)** resampling/simulation around TOPSIS is known; **(ii)** group TOPSIS with DM weighting (reliability/influence) is known; therefore, the novelty must be demonstrated as the *specific* combination *plus* a new robustness target and new evaluation/theory.

Exact matches, near-misses, and gaps

No exact keyword match found (terminology-level): the phrase "bagged fuzzy TOPSIS" is not established as a canonical method name in the mainstream literature (most hits are unrelated "bagging" usages). [5]

Near-miss cluster A: Bootstrap / resampling TOPSIS (very close conceptually)
- **Bootstrap sampling + fuzzy TOPSIS to produce a ranking distribution** for "large and distinct

decision makers" is explicitly proposed and demonstrated (including repeated runs and analysis of rank variability). This is very close to your "many TOPSIS runs" concept, even though their motivation is sampling variability rather than adversarial robustness. [6]
- **Bootstrap TOPSIS (crisp data)** has been used (e.g., flood risk contexts), showing that "TOPSIS + bootstrap" is a known design pattern. [7]
- Recent fuzzy TOPSIS variants use **bootstrap resampling of expert matrices** to compute confidence intervals and argue robustness/stability. Reviewers may treat your bagging as "a stronger form of bootstrap-based robustness analysis," unless you differentiate sharply. [8]
- A broader "compromise ranking with bootstrap confidence intervals" tradition also exists, showing that bootstrap is a recognised uncertainty quantification mechanism in decision/ranking contexts more generally. [9]

Near-miss cluster B: Monte Carlo + fuzzy TOPSIS for robustness
- A robust group decision framework combining **fuzzy TOPSIS with Monte Carlo simulation** has been published, explicitly motivated by avoiding bias introduced by naïve aggregation of linguistic/fuzzy inputs. [3]

Near-miss cluster C: DM reliability / weights in group (fuzzy) TOPSIS
- Group fuzzy TOPSIS literature explicitly discusses that simple averaging can mask discrepancy among DMs and motivates methods that preserve individual information or compute DM weights. [10]
- More broadly, fuzzy TOPSIS extensions for group decision making are well established, and a reviewer will expect you to cite foundational fuzzy TOPSIS and group TOPSIS extensions. [11]

Near-miss cluster D: "Random TOPSIS" naming collision risk
- "Random TOPSIS" already exists as a different concept (e.g., group modular random TOPSIS, GMC-RTOPSIS) focusing on heterogeneous information and "random/deterministic factors," not DM-bagging. If you use "Random TOPSIS" in your title, many reviewers will assume overlap unless you clearly disambiguate. [12]

Gap you can own (if executed rigorously):
The strongest gap is not "bootstrap TOPSIS exists," but: **a principled, contamination-aware ensemble framework for group fuzzy TOPSIS** that (1) targets *adversarial/low-quality DMs*, (2) links sampling/aggregation to robust statistics, and (3) provides formal stability guarantees + strong empirical validation. The ensemble MCDM literature explicitly notes the field is small and calls for clearer terminology, design choices, and evaluation frameworks—this is a favourable macro-context if you align closely with those expectations. [1]

Concrete, prioritised actions to strengthen novelty

1. **Rewrite the novelty claim around the *problem class*: "group fuzzy TOPSIS under adversarial contamination / biased DMs," not "bagging because Random Forest."** Anchor it to robust statistics language (contamination, outliers, breakdown), then show why existing bootstrap/Monte Carlo TOPSIS does not solve adversarial DM manipulation. [13]
2. **Differentiate from bootstrap-for-uncertainty papers** by formalising that your goal is *robust aggregation under corrupted agents*, not confidence intervals under sampling variability. Cite bootstrap-CI fuzzy TOPSIS papers as "uncertainty quantification baselines," not competitors. [14]
3. **Explicitly compare against the closest prior bootstrap-fuzzy-TOPSIS ranking-distribution method** and show: (i) different threat model, (ii) different aggregation target (final robust ranking vs distribution visualisation), (iii) additional reliability-aware mechanism. [2]
4. **Avoid "Random TOPSIS" in the title**; treat "Random" as overloaded. Use "bagged," "bootstrap-ensemble," "robust ensemble," or "contamination-aware." [12]

Specific references to cite (starting set, with identifiers)

- Chen-Tung Chen [15], “Extensions of the TOPSIS for group decision-making under fuzzy environment,” *Fuzzy Sets and Systems* (2000), doi:10.1016/S0165-0114(97)00377-1. [16]
- Ching-Lai Hwang & Kwangsun Yoon [17], *Multiple Attribute Decision Making* (TOPSIS origins), doi: 10.1007/978-3-642-48318-9_3. [18]
- Leo Breiman [19], “Bagging predictors” (1996), and “Random Forests” (2001), doi:10.1023/A:1010933404324. [20]
- Afful-Dadzie et al., “ranking distribution through bootstrap sampling… illustrated using fuzzy TOPSIS” (IEOM 2017). [2]
- Mojtahedi & Oo (2016), bootstrap TOPSIS for flood risk (crisp). [7]
- Vavatsikos et al. (2022), fuzzy TOPSIS + Monte Carlo in group decisions. [3]
- Bilal Bahaa Zaidan [21] et al. (2024), “An in-depth analysis of ensemble multi-criteria decision making…” *Applied Soft Computing*, doi:10.1016/j.asoc.2024.112267. [22]

Theoretical contribution

Assessment: Fail (must be strengthened for Q1).
Your report is rich in narrative, threat framing, and multiple algorithm variants (M1–M6), but it does not yet provide the kind of **formal model + propositions/lemmas** that high-tier reviewers expect when the claimed contribution is methodological and robustness-oriented. filecite turn0file0

What is missing (and what to add)

To be Q1-worthy, you do not need a fully general theorem for all fuzzy TOPSIS variants, but you should add **at least two** of the following:

1. **Formal problem definition (core):**
   Define a group fuzzy TOPSIS setting with (i) alternatives, criteria, fuzzy ratings (e.g., TFNs), weights, and (ii) a contamination model for DMs (e.g., an ε-fraction corrupted). A robust statistics contamination framing is standard and gives you a principled language to defend your “50% horizon” discussion. [23]

2. **Proposition on what bagging can/cannot do:**
   Bagging is classically a **variance reduction** technique, not a bias eliminator; this is well understood in ensemble learning. [20]
   A careful proposition such as: “Under i.i.d. honest DMs, bagging reduces variance of the estimated closeness coefficient; under systematic adversarial corruption, naïve uniform bagging does not guarantee bias reduction” will prevent reviewers from attacking your assumptions.

3. **Robust aggregation theorem/lemma linked to breakdown:**
   If your enhanced method relies on median-like or geometric-median-like consensus concepts, cite robust statistics results that the median/spatial median can reach a 50% breakdown point under contamination. [24]
   Then prove a *restricted* statement in your setting, e.g.: “If reliability scoring is based on an L1/spatial median of DM representations, then the reliability estimator is stable up to ε<0.5 contamination; hence sampling probabilities remain bounded away from attackers.”

4. **Stability/consistency guarantee for the ensemble rank:**
   Define your aggregation operator (Borda/mean/median of closeness) and prove a concentration or stability result (even if mild), supported by bootstrap theory foundations. [25]

Concrete, prioritised actions

1. Add a **definitions section** that introduces: decision tensor, DM, fuzzy rating type, distance, normalisation, and the ensemble estimator.
2. Add **one theorem on stability/variance** (bagging under i.i.d. ratings) and **one theorem/lemma on robustness** (under ε-contamination, with $\varepsilon<0.5$). [26]
3. Downgrade or remove absolute claims like "unbreakable until 50%" unless formally proven under explicit assumptions; otherwise state it as an empirical observation limited to your corruption model. filecite turn0file0

References to cite for theory framing

- Efron (1979), bootstrap foundations, doi:10.1214/aos/1176344552. [27]
- Peter J. Huber [28] , *Robust Statistics* (1981). [29]
- Robust median / spatial median breakdown point discussions. [24]

Methodology and algorithms

Assessment: Weak.
Your report contains multiple method variants (M1–M6) and gives many formulas, but Q1 reviewers will likely flag: (i) ambiguous baselines ("standard fuzzy TOPSIS" defined as min/mean/max aggregation may be contested), (ii) unclear hyperparameter guidance (B, K′, temperature), (iii) insufficient complexity analysis and reproducibility detail, and (iv) unclear separation between *threat model*, *estimator*, and *evaluation protocol*. filecite turn0file0

Concrete, prioritised actions

1. **Clarify the baseline definition (critical):**
   Explicitly cite which fuzzy TOPSIS formulation you implement (e.g., Chen 2000 vertex distance method) and which group aggregation operator you treat as baseline(s). [30]
   Then include **multiple** baseline aggregators (mean of TFNs, median/trimmed mean, OWA-style) so you are not criticised for a strawman baseline.

2. **Provide pseudocode that is implementable and unambiguous:**
   For each algorithm, specify:

3. input tensor shapes,
4. sampling procedure (with/without replacement),
5. aggregation operator,
6. output (full ranking vs top-1),

7. and the ensemble combiner.
   This is particularly important because ensemble MCDM reviews emphasise clear terminology, ensemble "levels," and evaluation design. [1]

8. **Complexity analysis:**
   Give big-O in terms of number of DMs (K), alternatives (m), criteria (n), bootstrap runs (B), and sample size (K′). This matters for your "up to 500 DMs" positioning. filecite turn0file0

9. **Hyperparameter rationale:**
   Provide principled defaults (e.g., $K' \approx 0.632K$ for bootstrap uniqueness, B chosen by convergence

of rank stability) and show sensitivity curves. Bootstrap foundations and ensemble-learning practice support such choices, but you must measure and report them. [31]

Key references to cite

- Chen (2000) for fuzzy TOPSIS under TFNs and distance. [16]
- Ensemble MCDM design and terminology review (Zaidan et al. 2024). [1]
- Bagging/Random Forest foundations (Breiman). [20]

Experiments, evaluation design, and baselines

Assessment: Fail (insufficient for Q1).
Your report includes synthetic datasets DS1–DS10 and multiple corruption rates, and it reports many “pass/fail” outcomes and rank changes across methods. filecite turn0file0 For Q1 journals, that is a promising start, but the experiment set is not yet convincing because it lacks: (i) standard real datasets with reproducible preprocessing, (ii) strong baselines from group decision and expert-weighting literatures, (iii) statistical testing, confidence intervals/effect sizes, and (iv) ablation studies isolating each design choice.

What Q1 reviewers typically demand in this area

1. **Real datasets (at least 2, ideally 3):**
   At minimum, include one “opinion survey / crowd rating” setting (large K), because bootstrap-fuzzy-TOPSIS ranking distribution work already uses that setting and sets expectations for external validity. [2]
   If biased data are not available, you can inject controlled corruptions, but you must still start from real distributions.

2. **Baselines beyond “standard fuzzy TOPSIS”:**
   You must compare against:

3. Monte Carlo fuzzy TOPSIS robustness baselines, [3]
4. bootstrap-CI style stability analysis papers (to show you go beyond CI), [8]
5. DM-weighting group TOPSIS methods (showing reliability weighting is not new alone), [10]
6. and at least one ensemble MCDM aggregation baseline (rank aggregation/voting). [1]

7. **Metrics and tests:**
   Rank correlation (Kendall τ, Spearman ρ) is fine, but any nonstandard indices you invent (DMII/CSI/BIR) must be mathematically defined, justified, and validated. filecite turn0file0
   Add statistical tests: permutation tests or paired bootstrap CIs over repeated corruptions/splits; report effect sizes and uncertainty. Bootstrap is a natural tool here and is well grounded. [32]

8. **Ablations (minimum set):**

9. no bagging (single aggregation),
10. bagging only,
11. bagging + robust aggregation (median/trim),
12. bagging + reliability weighting,

13. reliability weighting without bagging.
    This is essential to claim the reliability-aware variant is more than “expert weighting” in disguise. [33]

Concrete, prioritised actions

1. **Rebuild experiments around a formal corruption model** (ε-fraction adversarial; intensity parameter for how strongly they distort ratings). Tie this to robust statistics contamination framing. [34]
2. **Run $\geq$30 random seeds per setting** and report mean ± 95% CI for τ/ρ and top-1 accuracy. Use paired tests for method comparisons. [32]
3. **Add “closest prior art” baselines** explicitly: (i) bootstrap ranking distribution fuzzy TOPSIS, (ii) Monte Carlo fuzzy TOPSIS, (iii) DM-weighted group TOPSIS. [35]
4. **Visualisations expected for robustness papers:**
5. stability curves vs ε (corruption fraction),
6. sensitivity vs K' and B,
7. rank-frequency histograms across bags,
8. and confidence intervals for closeness coefficients. [36]

Results and interpretation

Assessment: Weak.
Your report claims sharp thresholds (notably a “50% horizon”) and presents multiple dataset scenarios where methods succeed/fail. filecite turn0file0 The main issue is that the evidence is not yet presented with appropriate statistical uncertainty, and the conclusions are stronger than what the current design can support.

Concrete, prioritised actions

1. **Replace binary “success/failure” narration with probabilistic robustness curves:**
   e.g., P(correct top-1), median Kendall τ, etc., with confidence bands. Bootstrap CIs are suitable and consistent with the theme. [32]
2. **Explain why bagging alone should not defeat bias** (variance vs bias distinction) and show empirically where it helps (variance reduction) versus where reliability weighting is necessary (bias control). [20]
3. **Report effect sizes** (e.g., $\Delta\tau$ and Δtop-1 accuracy) and perform “robustness to attacker strength” stress tests, not just attacker fraction.

References to cite for interpretation framing

- Bagging and Random Forest basics: why bagging stabilises (variance) rather than guarantees bias removal. [20]
- Robust statistics breakdown and contamination framing to interpret the “50%” phenomenon correctly. [37]

Writing, positioning, and naming

Assessment: Weak.
The report is written as an “architectural” manifesto with very strong claims and security-style rhetoric,

and it sometimes frames “standard fuzzy TOPSIS” in a way that could be contested. fileciteturn0file0
For Q1 journals, tone and positioning must be tightened:

- Use precise, restrained claims.
- Separate *threat model* from *method design* and *evidence.*
- Avoid overloaded terms (“Random TOPSIS”). 12

Concise contribution paragraph (template)

This is a Q1-style paragraph you can adapt:

> We study multi-criteria group decision making with linguistic/fuzzy evaluations in the presence of biased or low-quality decision makers. Building on fuzzy TOPSIS, we propose an ensemble estimator that repeatedly resamples decision makers, performs fuzzy TOPSIS on each bootstrap panel, and aggregates the resulting rankings/closeness coefficients into a final robust ranking. We further introduce a reliability-aware extension in which decision-maker sampling and aggregation weights are learned from consensus-consistency signals, yielding a contamination-aware framework. We provide theoretical results characterising stability and robustness under an ε-contamination model and conduct extensive experiments on real and synthetic datasets, including controlled adversarial bias injection, demonstrating improved ranking stability and bias resistance relative to existing fuzzy TOPSIS group aggregation, Monte Carlo/bootstrapped variants, and decision-maker weighting baselines.

This paragraph is defensible if—and only if—you deliver the theory + experiments above. 38

Title options that reduce confusion

Because “Random TOPSIS” already denotes other methods (e.g., GMo-RTOPSIS), avoid that phrase. 39

- “A Bootstrap-Ensemble Framework for Robust Group Fuzzy TOPSIS under Decision-Maker Contamination”
- “Bagged Group Fuzzy TOPSIS with Reliability-Aware Decision Maker Sampling”
- “Contamination-Aware Ensemble Fuzzy TOPSIS for Large-Scale Group Decision Making”
- “Robust Ensemble Aggregation for Group Fuzzy TOPSIS: Theory and Adversarial Evaluation”

Key references to cite for positioning

- “Random TOPSIS” existing terminology: Lourenzutti & Krohling (GMo-RTOPSIS; GMC-RTOPSIS). 12
- Ensemble MCDM taxonomy and “ensemble levels” framing: Zaidan et al. (2024). 1

Publication strategy and revision plan

Target Q1 journal shortlist and fit

Below are five strong Q1 targets aligned with (robust) MCDM + soft computing + decision analytics. These are demanding venues; they will require a fully reproducible experimental package and a crisp theoretical story.

Journal	Fit for your topic	Impact Factor / CiteScore (publisher-reported)	Typical decision times (publisher-reported)	Notes
Expert Systems with Applications [40]	Excellent for fuzzy/ ensemble decision methods with strong experiments	IF 7.5; CiteScore 15.0 [41]	~5 days to first decision; ~147 days to acceptance [41]	Needs clear novelty beyond "another fuzzy TOPSIS variant"
Applied Soft Computing [42]	Excellent if you emphasise robustness + formalism + code	IF 6.6; CiteScore 14.5 [43]	(Use journal reporting; focus on strong revision quality) [43]	Also published major ensemble MCDM review—good context but higher bar [1]
Information Sciences [44]	Strong for methodological contributions + theory + experiments	IF 6.8; CiteScore 14.4 [45]	~2 days to first decision; ~170 days to acceptance [45]	Reviewers will demand rigorous baselines and proofs
Decision Support Systems	Strong fit if framed as decision support + evaluation rigor	IF 6.8; CiteScore 13.5 [46]	~2 days to first decision; ~285 days to acceptance [47]	Requires careful theory validation and real decision-support relevance [48]
Omega [49]	High prestige; will require very strong contribution	IF 7.2; CiteScore 14.9 [50]	~3 days to first decision; ~244 days to acceptance [50]	Hardest among these; must look like a generalisable OR method

Quartile confirmation (library-reported JCR-JIF/SJR): Applied Soft Computing, Information Sciences, Omega, and Decision Support Systems are shown as Q1 in the referenced listings. [51] For Expert Systems with Applications, a library listing indicates Q1 status and high CiteScore percentile. [52]

Expected reviewer concerns (what to pre-empt)

1. **"This is just bootstrap/Monte Carlo TOPSIS again."** You must explicitly contrast with prior bootstrap ranking-distribution and Monte Carlo fuzzy TOPSIS work. [53]
2. **"Bagging reduces variance, not adversarial bias."** Address this with theory + experiments; do not oversell bagging alone. [20]
3. **"DM weighting is already known."** Show that your reliability-aware mechanism is distinct and that *the ensemble* + *contamination model* + *evaluation* is the real contribution. [33]
4. **Reproducibility and baselines:** Q1 venues increasingly expect code/data availability and thorough baseline comparisons. [54]

Prioritised revision checklist

High priority (must-have): - Full related-work rewrite covering: bootstrap TOPSIS, Monte Carlo fuzzy TOPSIS, DM-weighting group TOPSIS, ensemble MCDM taxonomy. [55]

- Formal definitions + at least 2 theoretical statements (stability + robustness). [56]
- Real datasets + rigorous synthetic corruption; ≥30 runs; CIs; statistical tests; ablations. [14]
- Reproducible code release checklist (below).

Medium priority (strongly recommended): - Complexity analysis and hyperparameter sensitivity. [57]
- Cleaner naming and threat model formalisation; remove extreme rhetoric. [39]

Reproducibility checklist (minimum for Q1)

- Public code repository with tagged release (v1.0) and fixed random seeds.
- Exact implementation details of fuzzy TOPSIS variant (distance, normalisation, defuzzification if any). [16]
- Data preprocessing scripts + corruption injection scripts.
- Environment file (conda/pip), dependency versions, hardware notes.
- "One-command" reproduction for all tables/figures.
- Clear licence and citation instructions.

Revision timeline (Gantt-style, Mermaid)

```
gantt
    title BFTOPSIS to Q1-ready manuscript (12-week plan)
    dateFormat  YYYY-MM-DD
    axisFormat  %d %b

    section Positioning & Literature
    Full related-work rebuild (bootstrap/MC/DM-weighting/ensemble MCDM) :a1, 2026-03-08, 14d
    Threat model + contribution reframing (contamination-aware robustness) :a2, after a1, 7d

    section Theory
    Formal definitions + notation + estimator specification :b1, after a2, 7d
    Stability/robustness propositions + proof sketches :b2, after b1, 14d

    section Implementation
    Reference implementation + baselines + unit tests :c1, 2026-03-15, 21d
    Reproducibility packaging (configs, scripts, docs) :c2, after c1, 7d

    section Experiments
    Real datasets + synthetic corruption suite :d1, 2026-03-29, 21d
    Statistical analysis + ablations + plots :d2, after d1, 14d

    section Writing
    Results/discussion rewrite + limitations + appendix :e1, 2026-04-26, 14d
    Final polish, journal formatting, response-to-reviewer prewrite :e2, after e1, 7d
```

Bottom line

- **Is it novel enough today for Q1?** Not yet—because close prior work exists for bootstrap/Monte Carlo fuzzy TOPSIS workflows and DM-weighted group TOPSIS, and your manuscript currently lacks the theory + experimental rigor needed to defend the "Q1-worthy" novelty claim. [55]
- **Can it be made Q1-worthy?** Yes—if you (1) **reframe** it as contamination-aware robust group fuzzy TOPSIS, (2) **differentiate** explicitly from bootstrap-for-uncertainty and Monte Carlo robustness papers, (3) add **formal definitions + at least two theoretical results**, and (4) execute **multi-dataset, statistically rigorous, reproducible experiments** with strong baselines and ablations. [58]

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[1] [22] [38] [40] [57] [58] https://www.sciencedirect.com/science/article/abs/pii/S156849462401041X
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